

University Interscholastic League
2024 – 2025 Elementary Number Sense Test B

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

- | | |
|---|---|
| <p>(1) $24 + 45 =$ _____</p> <p>(2) $78 + 23 =$ _____</p> <p>(3) $224 - 56 =$ _____</p> <p>(4) $918 \div 3 =$ _____</p> <p>(5) $5 \times 48 =$ _____</p> <p>(6) $402 - 127 =$ _____</p> <p>(7) $36 + 16 + 14 =$ _____</p> <p>(8) $64 \times 11 =$ _____</p> <p>(9) $50 \times 23 =$ _____</p> <p>*(10) $1 + 897 + 1203 =$ _____</p> <p>(11) $16 \times 25 =$ _____</p> <p>(12) 38670.4782 rounded to the hundreds place is
_____</p> <p>(13) Which digit is in the thousands place in
34917.20568? _____</p> <p>(14) $31 \times 29 =$ _____</p> <p>(15) What is the remainder for $46188 \div 9$? _____</p> <p>(16) The number of odd whole numbers between 7 and
19 is _____</p> <p>(17) $5 \times 10^{-1} + 6 \times 10^1 + 2 \times 10^{-3} =$ _____ (decimal)</p> <p>(18) $57 \times 15 - 57 \times 5 =$ _____</p> <p>(19) XCIV = _____ (Arabic Numeral)</p> | <p>*(20) $667 \times 39 =$ _____</p> <p>(21) $18 + 24 \div 2 \times 3 =$ _____</p> <p>(22) $34 + 30 + 26 + 22 =$ _____</p> <p>(23) $\frac{5}{12}$ hour = _____ minutes</p> <p>(24) $143 \times 7 =$ _____</p> <p>(25) $\frac{19}{24} - \frac{3}{24} =$ _____</p> <p>(26) $102 \times 101 =$ _____</p> <p>(27) 76% = _____ common fraction</p> <p>(28) If 15 ♥ costs 27¢, then 10 ♥ cost _____ ¢</p> <p>(29) $28 \div \frac{7}{10} =$ _____</p> <p>*(30) $126 \times 721 + 54 =$ _____</p> <p>(31) \$39.75 = _____ quarters</p> <p>(32) The smallest prime number between 60 and 50 is
_____</p> <p>(33) 64 ounces = _____ pints</p> <p>(34) $44 \times 18 =$ _____</p> <p>(35) $87\frac{1}{2}\%$ = _____ common fraction</p> <p>(36) The LCM of 24 and 15 is _____</p> <p>(37) $36 \times 12 =$ _____</p> |
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- (38) If 12 apples cost \$3.60, then 8 apples cost \$_____
- (39) 12 is to 20 as 4 is to _____
- *(40) $161 \times 625 =$ _____
- (41) $\frac{5}{16} + \frac{7}{16} =$ _____ (common fraction)
- (42) $14^2 - 4 =$ _____
- (43) If $f(x) = 2x + 7$, then $f(12) =$ _____
- (44) The diameter of a circle with an area of 144π is

- (45) $3^3 \times 3^0 =$ _____
- (46) $2\frac{1}{8} \times 2\frac{7}{8} =$ _____ (mixed number)
- (47) What is the perimeter of a square with area 64? _____
- (48) $77\frac{7}{9}\% \times 36 =$ _____
- (49) If there are 128 grapes to a pound, then 1 ounce equals how many grapes? _____
- *(50) $334\frac{1}{70} \times 359 =$ _____
- (51) What is the number, **k**, in the sequence:
0, 3, 8, **k**, 24, 35, . . . ? _____
- (52) $4\frac{3}{4} + 5\frac{7}{10} =$ _____ (mixed number)
- (53) $\sqrt[3]{64} =$ _____
- (54) A square with area 16 is located inside a square with perimeter 24. What is the area between the squares?

- (55) 24 (base 6) = _____ (base 10)
- (56) The number of elements in $\{1, 2, 4, 6\} \cap \{1, 2, 3, 4\}$ is _____
- (57) What is the perimeter of a regular octagon with side $4\frac{1}{2}$? _____
- (58) $75 \times 18 =$ _____
- (59) $12\frac{7}{9} - 3\frac{1}{3} \times 3\frac{1}{3} =$ _____
- *(60) 200 inches = _____ centimeters
- (61) 31 (base 4) = _____ (base 2)
- (62) $(-225) \times (-4) =$ _____
- (63) What is the probability a getting a sum of 3 when rolling a pair of dice? _____
- (64) 4 quarts = _____ pints
- (65) $48^2 =$ _____
- (66) $12^5 \div 8$ has remainder of _____
- (67) The area of a rhombus with diagonals 10 and $12\frac{1}{2}$ is

- (68) If $-3x - 15 < -12$, then $x >$ _____
- (69) $(-40) \div (-0.25) =$ _____
- *(70) $19^3 =$ _____
- (71) The multiplicative inverse of $-3\frac{3}{8}$ is _____
- (72) $18^2 - 32^2 =$ _____
- (73) If 150% of 220 = _____
- (74) What is the area of a trapezoid with bases $5\frac{1}{8}$, $9\frac{1}{8}$ and altitude to the bases of 8? _____
- (75) $15^2 + 30^2 =$ _____
- (76) $(1 + 2 + 3 + \dots + 10)^2 =$ _____
- (77) What is the total surface area of a rectangular box with edges 4, 7 and 6? _____
- (78) The area of a right triangle with legs $8\frac{2}{3}$ and 12 is

- (79) $625 \times 80 =$ _____
- *(80) 25 square miles = _____ acres

2024 – 2025 University Interscholastic League Elementary Number Sense Test B – Key

(1) 69	*(20) 24713 – 27313	(38) 2.40	(59) $1\frac{2}{3}; \frac{5}{3}$
(2) 101	(21) 54	(39) $6\frac{2}{3}; \frac{20}{3}$	*(60) 483 – 533
(3) 168	(22) 112	*(40) 95594– 105656	(61) 1101
(4) 306	(23) 25	(41) $\frac{3}{4}$	(62) 900
(5) 240	(24) 1001	(42) 192	(63) $\frac{1}{18}$
(6) 275	(25) $\frac{2}{3}$	(43) 31	(64) 8
(7) 66	(26) 10302	(44) 24	(65) 2304
(8) 704	(27) $\frac{19}{25}$	(45) 27	(66) 0
(9) 1150	(28) 18	(46) $6\frac{7}{64}$	(67) $62\frac{1}{2}; \frac{125}{2}; 62.5$
*(10) 1996 – 2206	(29) 40	(47) 32	(68) -1
(11) 400	*(30) 86355 – 95445	(48) 28	(69) 160
(12) 38700	(31) 159	(49) 8	*(70) 6517 – 7201
(13) 4	(32) 53	*(50) 113916 – 125906	(71) $-\frac{8}{27}$
(14) 899	(33) 4	(51) 15	(72) -700
(15) 0	(34) 792	(52) $10\frac{9}{20}$	(73) 330
(16) 5	(35) $\frac{7}{8}$	(53) 4	(74) 57
(17) 60.502	(36) 120	(54) 20	(75) 1125
(18) 570	(37) 432	(55) 16	(76) 3025
(19) 94		(56) 3	(77) 188
		(57) 36	(78) 52
		(58) 1350	(79) 50000
			*(80) 15200 – 16800

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.